

August 21, 2026

Dear WSP Hiring Team,

I am writing to apply for the Early Professional, Geographic Information Systems Science position with WSP in Calgary. With a Master of Science in Computer Science from the University of Calgary and hands-on experience building geospatial data systems, remote-sensing workflows, and Python/C++ processing tools, I am excited by the opportunity to support WSP's environmental and engineering teams through reliable GIS analysis and spatial data products.

My graduate research focused on scalable management of Earth observation data. As a Research Software Developer with BigGeo/Mitacs at the University of Calgary, I built C++ and Python software for processing, storing, and retrieving large satellite-imagery datasets. I automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs, and developed DGGS-based spatial encoding workflows that transformed imagery into structured tensors for multiresolution retrieval and analysis.

This background aligns strongly with GIS work that depends on careful data manipulation, QA/QC, documentation, and collaboration with technical project teams. In my thesis work, I designed spatial data structures, assessed storage and retrieval performance, and communicated results through a first-author publication and research presentations. I also developed a digital elevation model super-resolution framework using Python, PyTorch, Rasterio, Google Earth Engine, Streamlit, and 3D visualization tools, which strengthened my ability to evaluate spatial outputs and explain technical trade-offs clearly.

I also bring backend and data-engineering experience that would help me contribute beyond map production alone. In a previous Back-End Developer role, I built Django REST APIs backed by PostgreSQL, implemented authentication and service-integration logic, and worked with Git-based development workflows. I have also mentored students in AWS-based data pipelines, large-scale data processing, debugging, and database coursework. While I would continue building deeper production experience with ESRI tools, ArcGIS Online/Enterprise, and cartographic standards, my foundation in Python, SQL, spatial data, automation, and documentation is directly relevant to this role.

I am particularly interested in WSP because the role sits at the intersection of geospatial technology, environmental work, and practical client-facing analysis. I would be grateful for the opportunity to contribute to GIS products that help project teams understand places, risks, and environmental decisions more clearly.

Thank you for considering my application. I look forward to the opportunity to discuss how my geospatial software, remote-sensing, data-processing, and documentation experience can contribute to WSP.

Sincerely,
Amir Mirzai Golpayegani
amirgolpaa24@gmail.com